

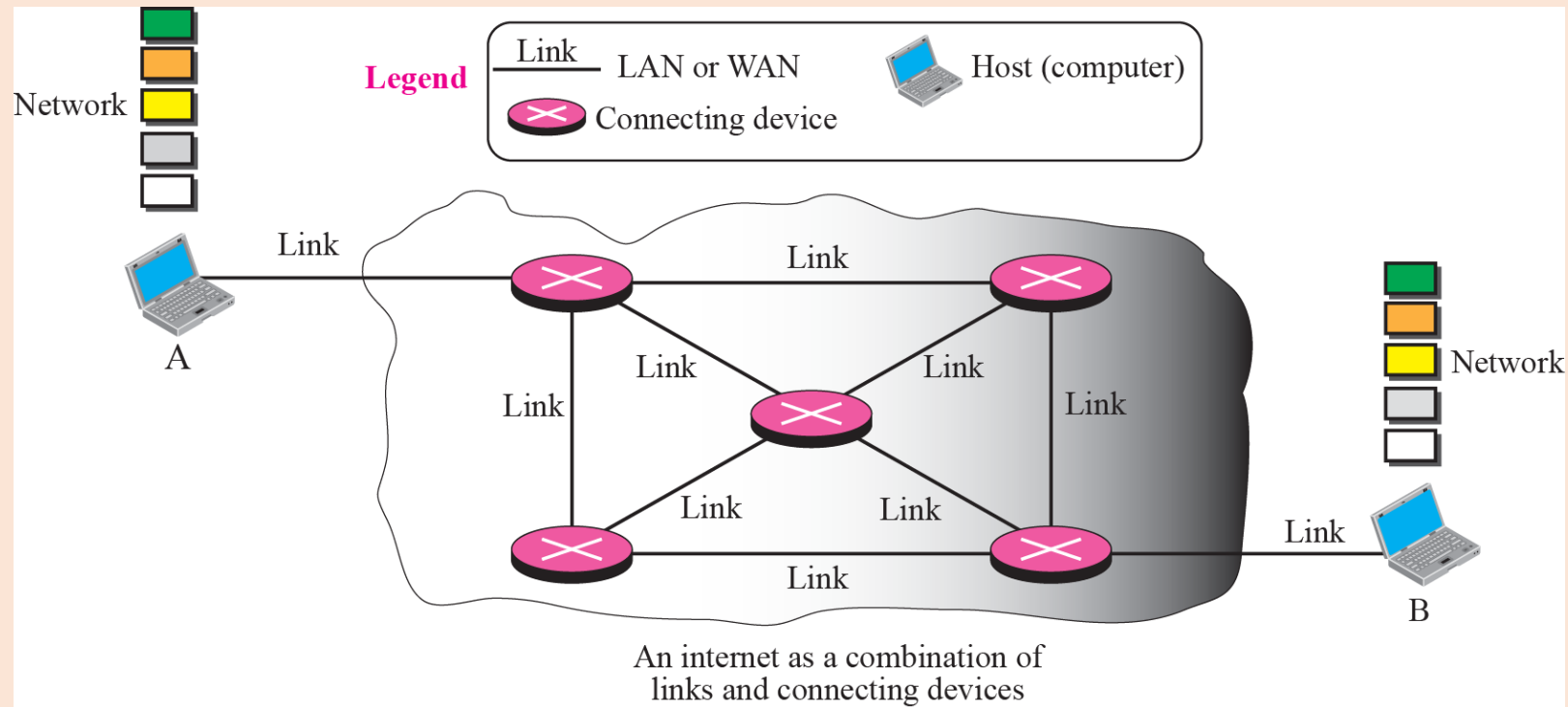
Introduction to Network Layer

Reference:

Behrouz A. Forouzan, TCP/IP Protocol Suite, 4th Edition, Tata McGraw Hill, 2010

The Internet

- Not one single network
- Made of many networks connected together through the connecting devices.
- So inter-network

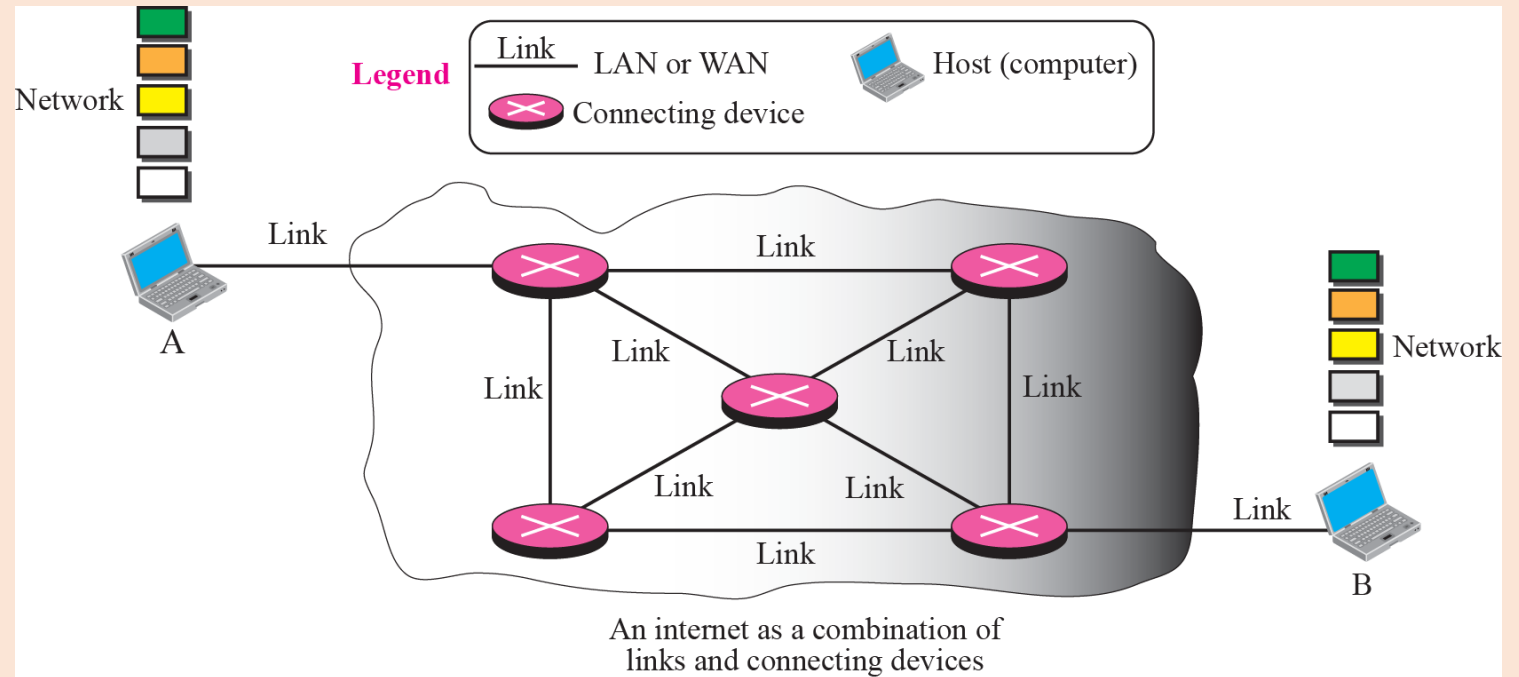
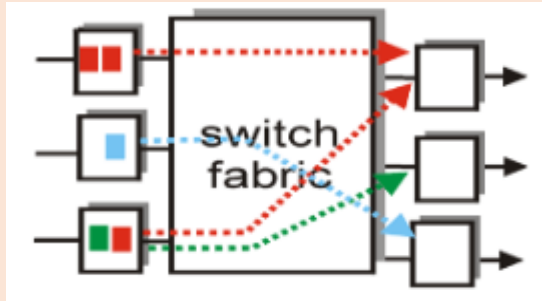


An imaginary small internet with a few networks and a few connecting devices

How data flows from A to B ? - Role of Routers. Switching Process

SWITCHING

When a message reaches a connecting device, a **decision needs to be made** to select one of the output ports through which the packet needs to be send out.



Connecting device **acts as a switch** that **decides & connects** one port to another port.
This process is known as - **Switching**

TYPES OF SWITCHING

Following are the two solutions to achieve Switching.

- ✓ **Circuit Switching**
- ✓ **Packet Switching**

Circuit Switching

A physical circuit (or channel) is established between the source and destination of the message before the delivery of the message.

- whole message is sent
- no need to divide into smaller chunks or packets

After transmission, circuit (or channel) is established is removed.

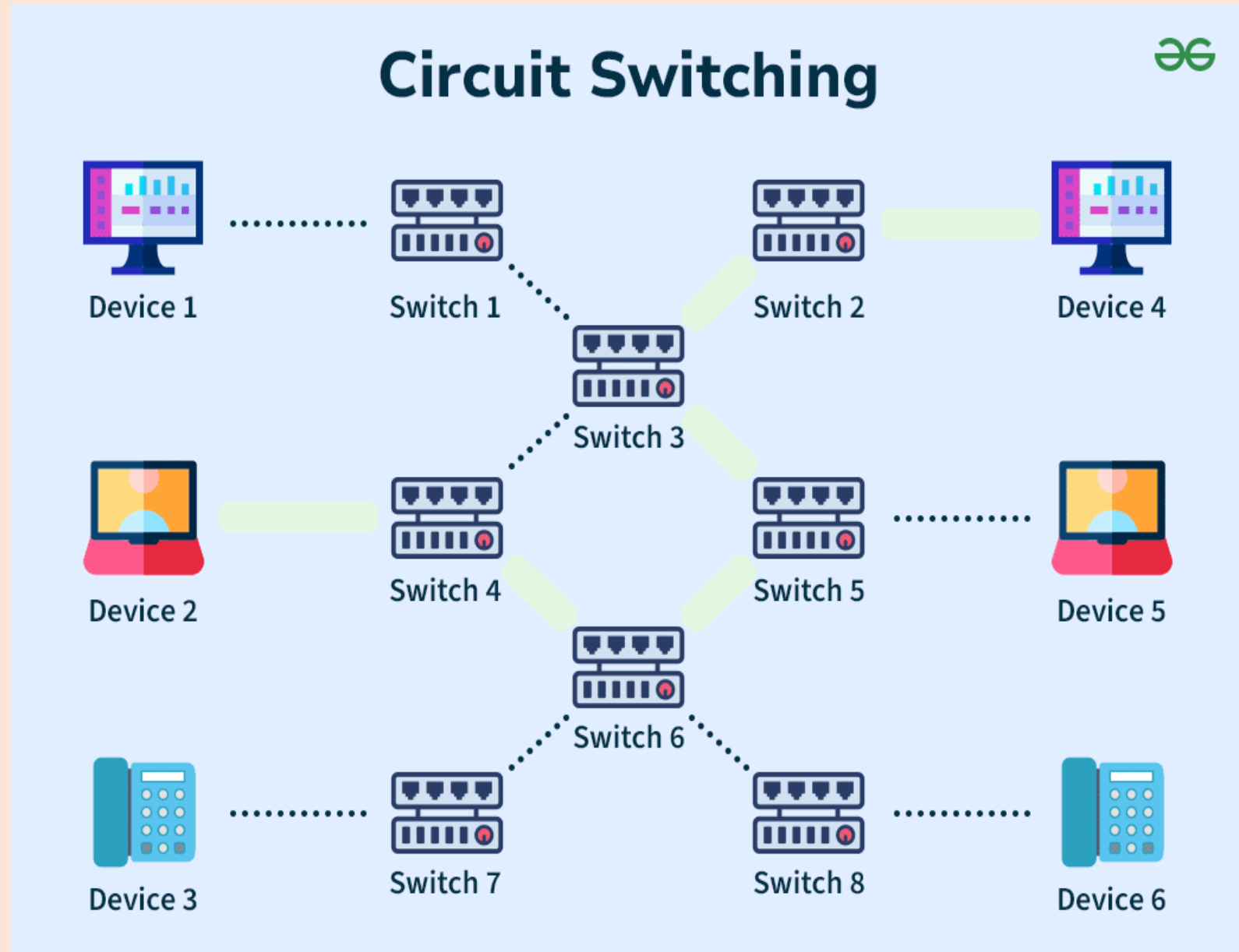
Example-

A good example of a circuit-switched network is the **early telephone systems** in which the path was established between a Caller and a Callee when the telephone number of the Callee was dialed by the Caller.

Circuit Switching

Initially physical circuit (or channel) is established between the source and destination

After transmission, circuit (or channel) is removed.



Source: <https://www.geeksforgeeks.org/computer-networks/circuit-switching-in-computer-network/>

Packet Switching

The **network layer in the Internet today** is a packet-switched network.

- The message is first **divided into manageable chunks** or packets at Source(upper layers) & transmitted.
- The Routers need to decide **how to route** the packets to the final destination.
- The packets are **assembled at the destination**.

Today, a packet-switched network can use two approaches-

- ✓ **Datagram Approach** -independent path (Connectionless)
- ✓ **Virtual Circuit Approach** – a fixed path (Connection Oriented)

PACKET SWITCHING...

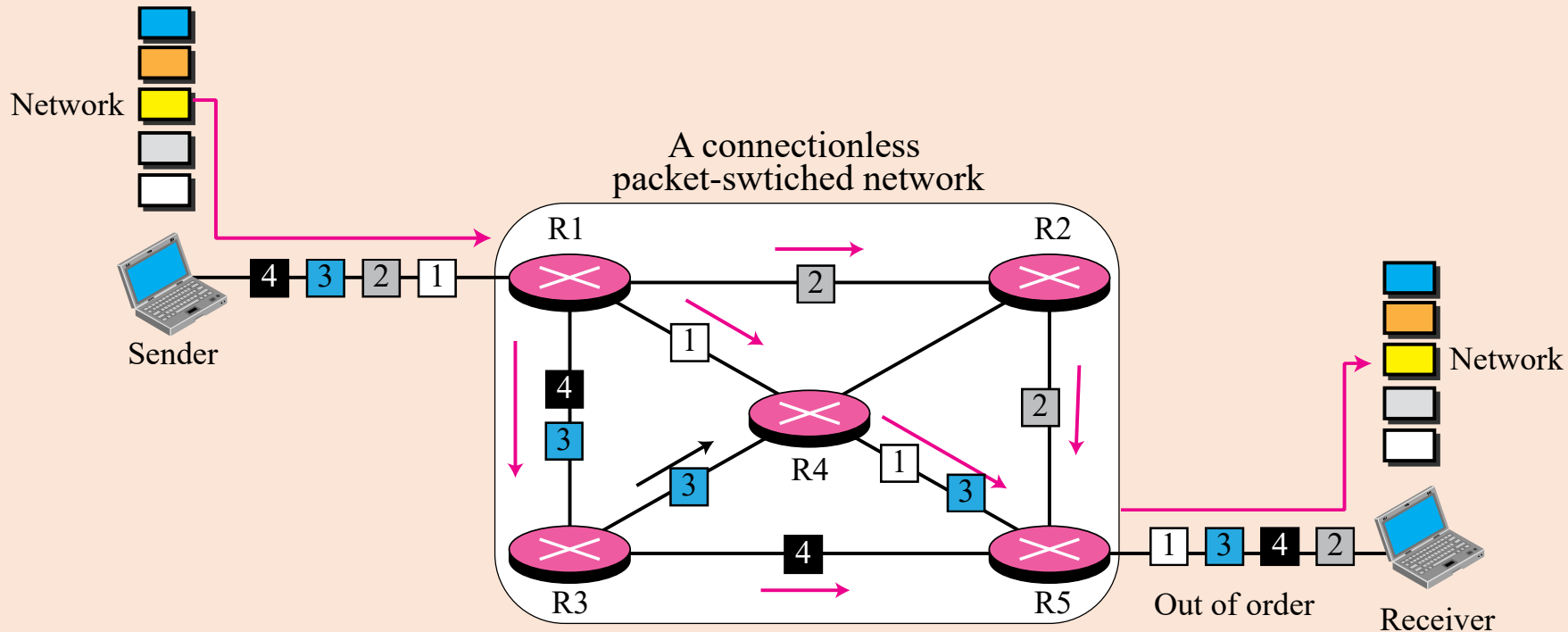
Datagram Approach

- The manageable chunks or packets are called as Datagrams
- - The packet-switched network layer of the Internet was designed as-
 - Connectionless Service but tendency is towards Connection oriented

PACKET SWITCHING...

a. Connectionless Service

- The **manageable chunks** or packets are called as **Datagrams**
- The network layer protocol **treats each packet independently**.
- Each packet has **no relationship** to any other packet.
- Datagram of a message may or **may not** travel the **same path**.



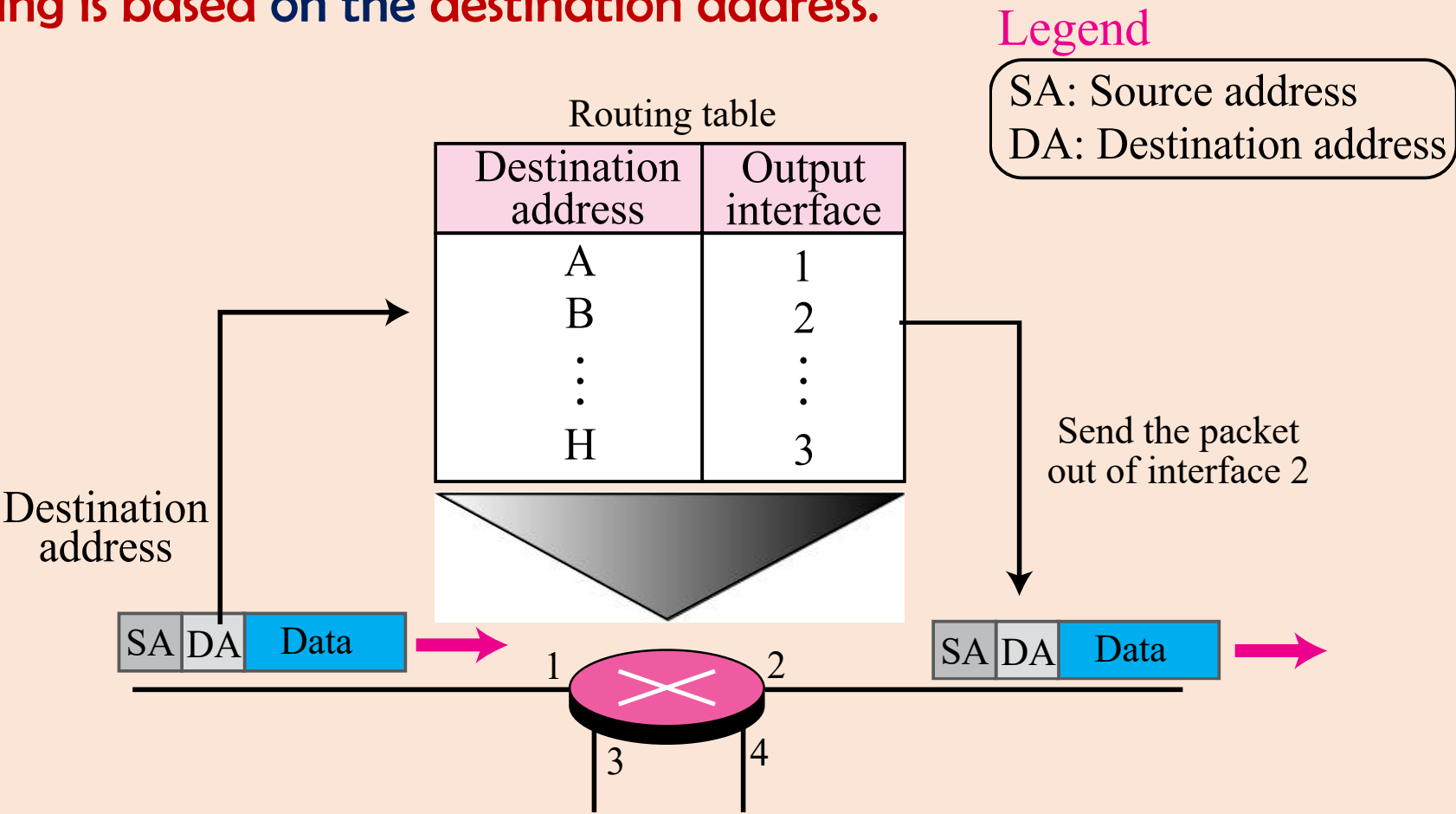
- Initially to keep **design simple**, **Internet** was decided to make **Connectionless Service**

PACKET SWITCHING...

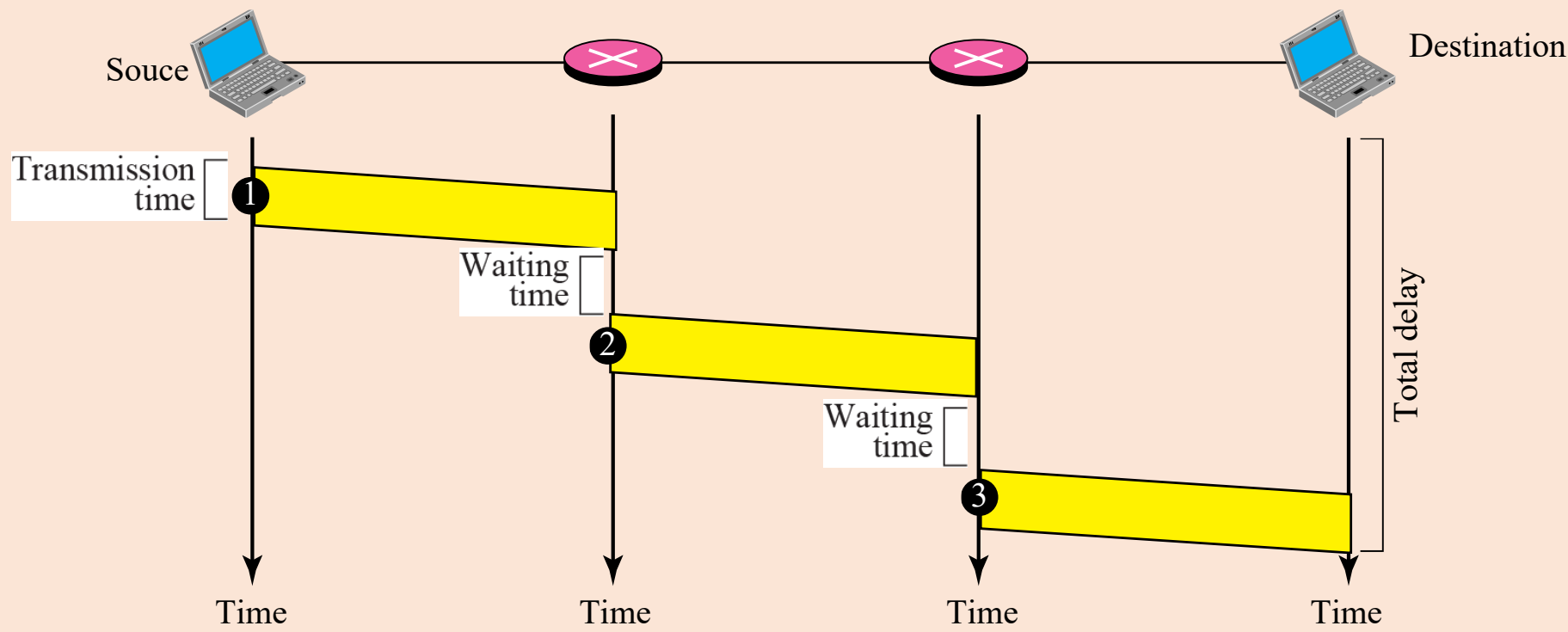
Forwarding process in **connectionless Service Network**

The packet contained in its header: **source**, **destination address** & other information.

Packet **routing** is based on the **destination address**.



Delay in Connection less Network



PACKET SWITCHING...

b. Connection Oriented Service

- The **manageable chunks** or packets are called as **Datagrams**
- There is a **relation between all packets** belonging to a message.
- A **virtual connection** should be set up to define the path for the datagrams.

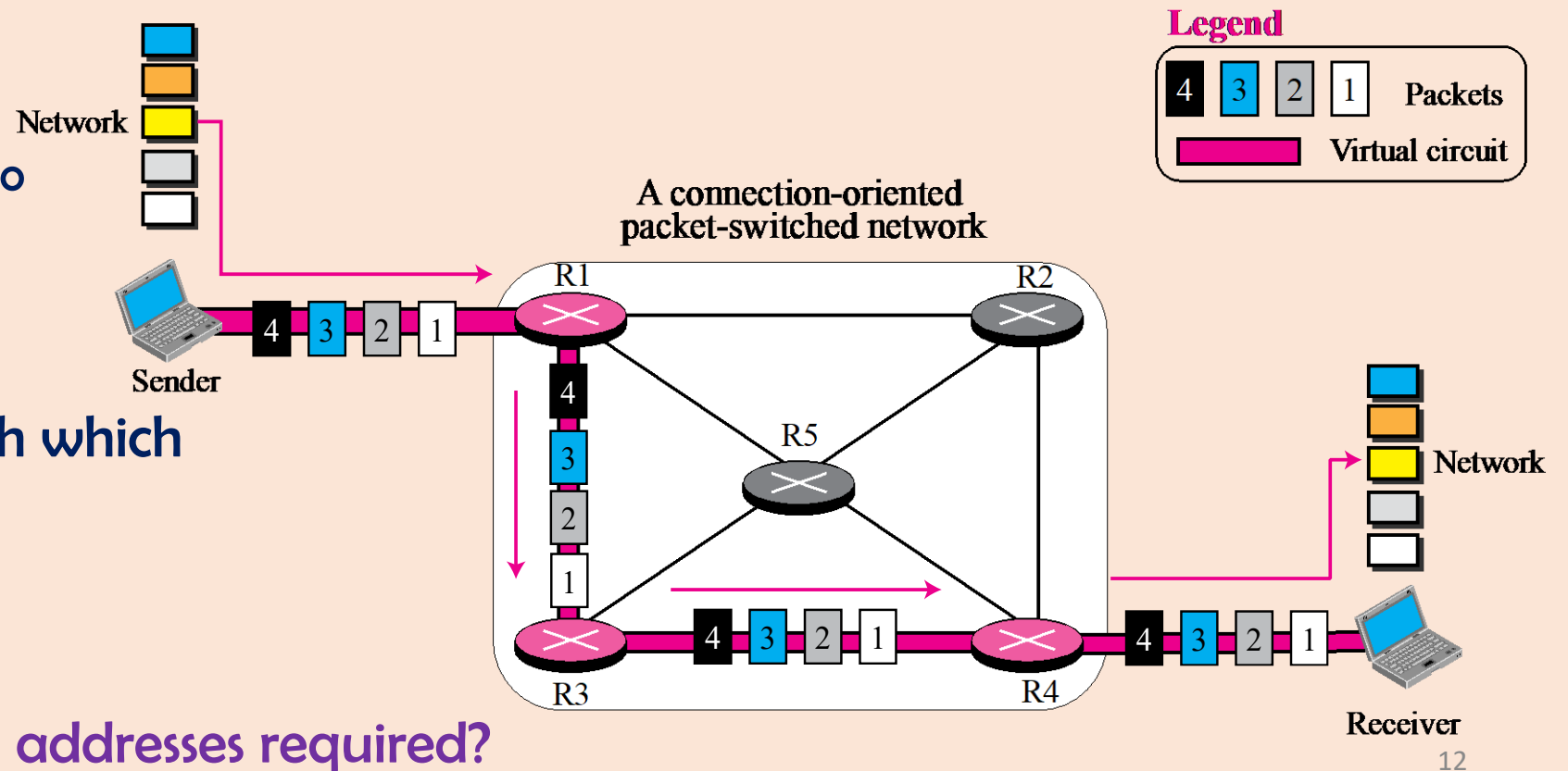
Packet contains-

Source , Destination Address also
Virtual Circuit identifier- VCI

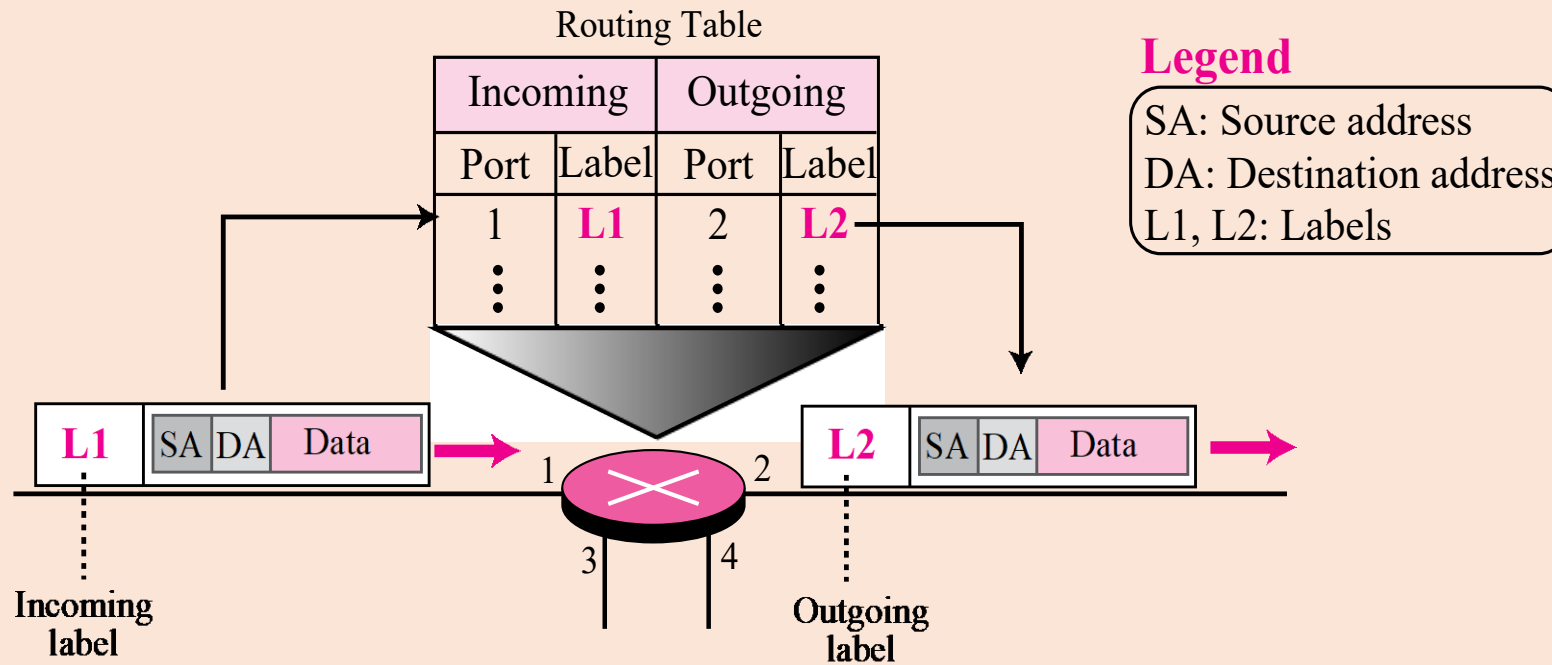
VCI determines path through which
packet must pass through.

Is only **VCI** enough ?

Is still **Source, Destination IP** addresses required?



Forwarding process in a connection-oriented network



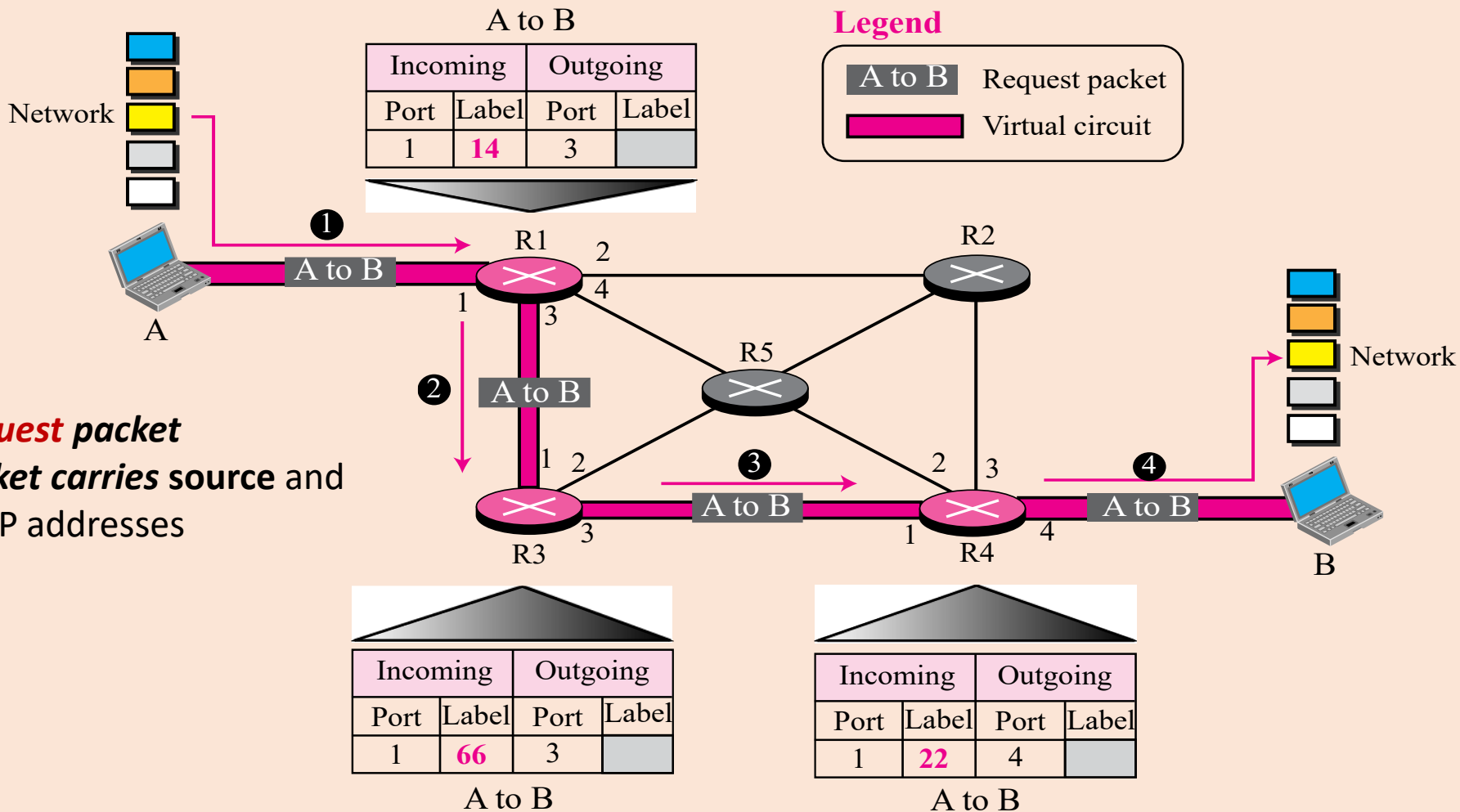
How **Labels entries** are made into the Routing table ?

PACKET SWITCHING...

Virtual Circuits -

A **three-phase** process is used: *setup*, *data transfer*, and *teardown*

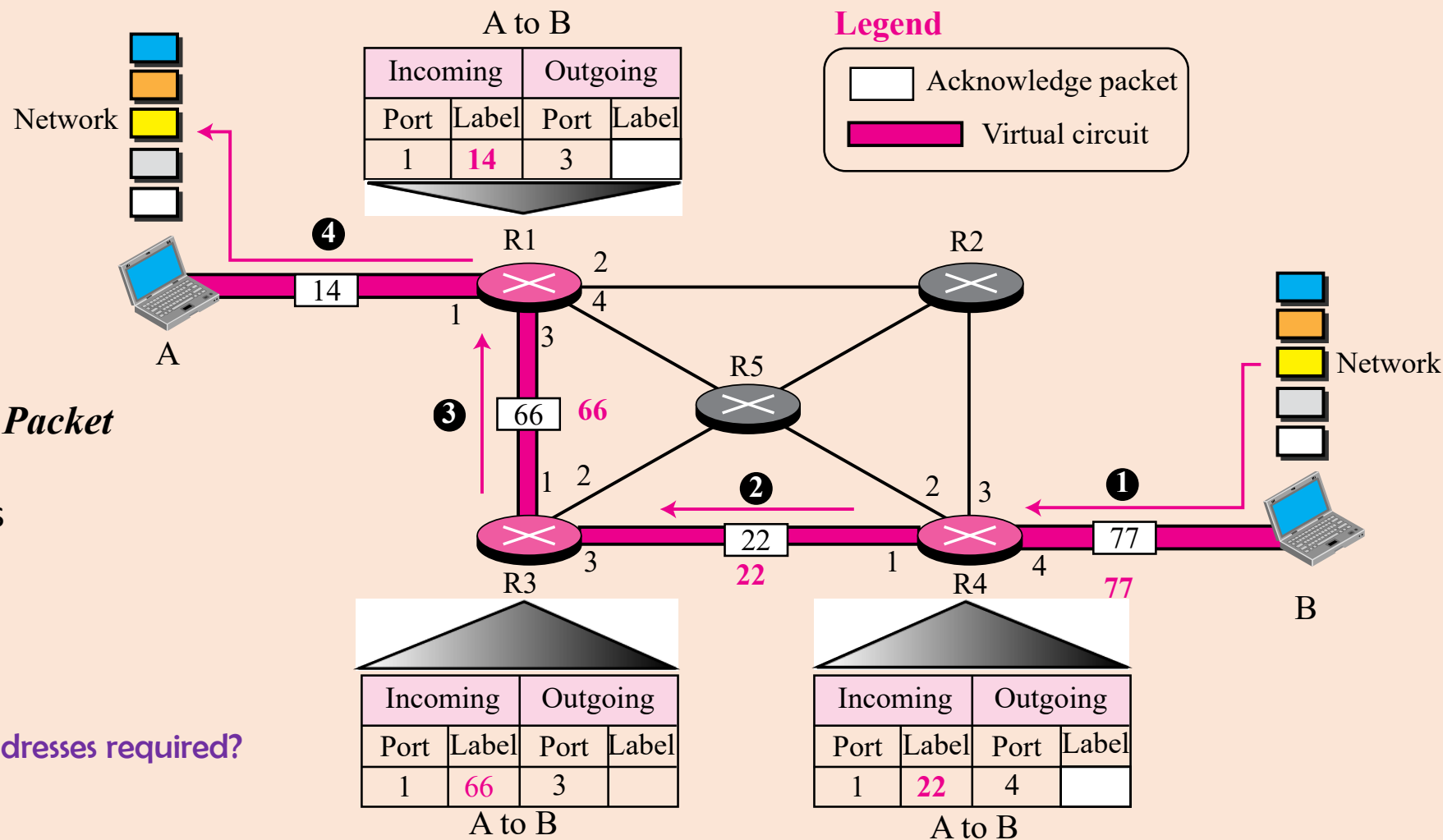
Setup Phase uses two auxiliary packets named – **Request** & **Acknowledgement** packets.



PACKET SWITCHING...

Virtual Circuits -

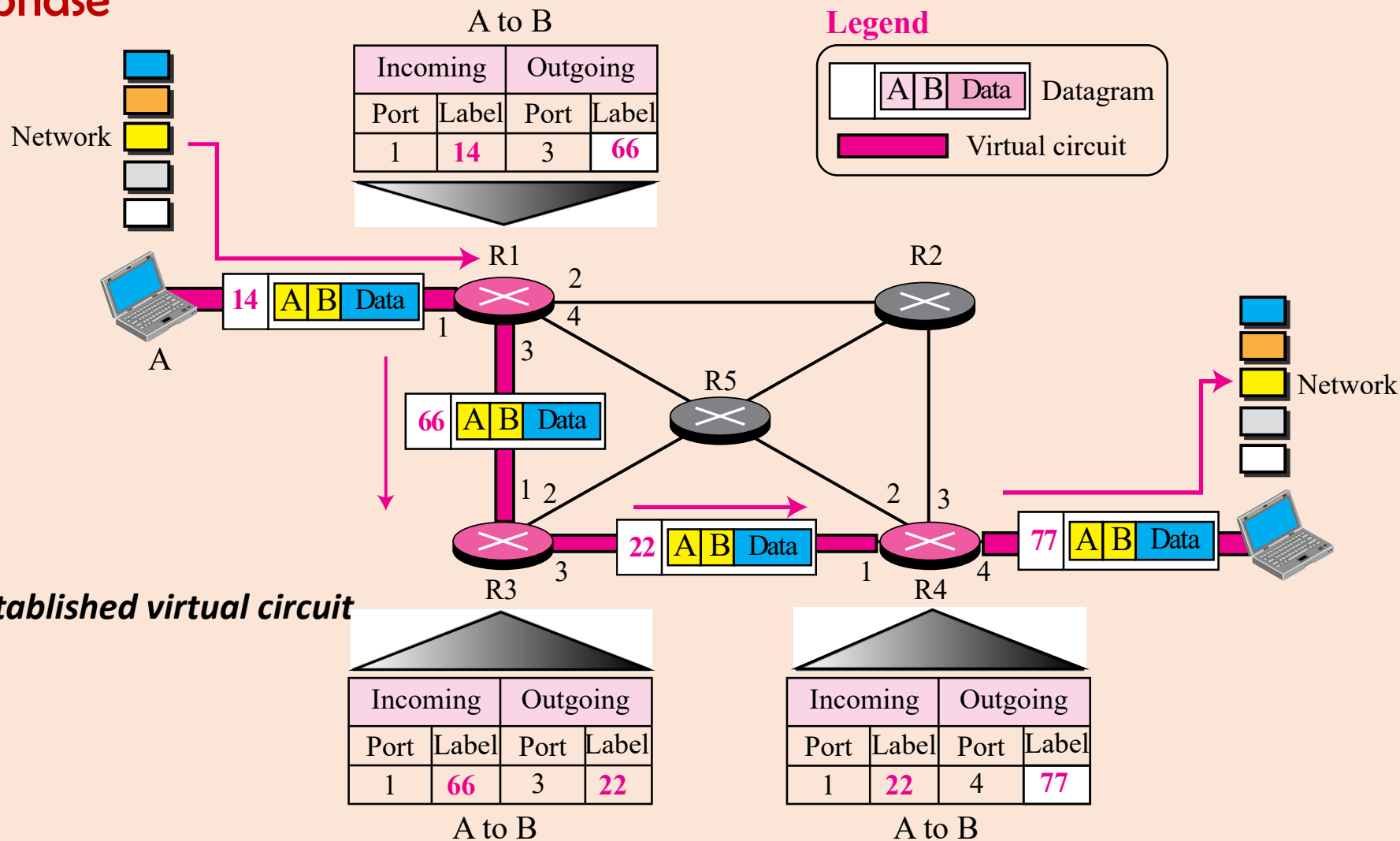
Setup Phase uses two auxiliary packets named – **Request** & **Acknowledgement** packets.



PACKET SWITCHING...

Virtual Circuits -

Data Transfer phase



Virtual Circuits -

Teardown Phase

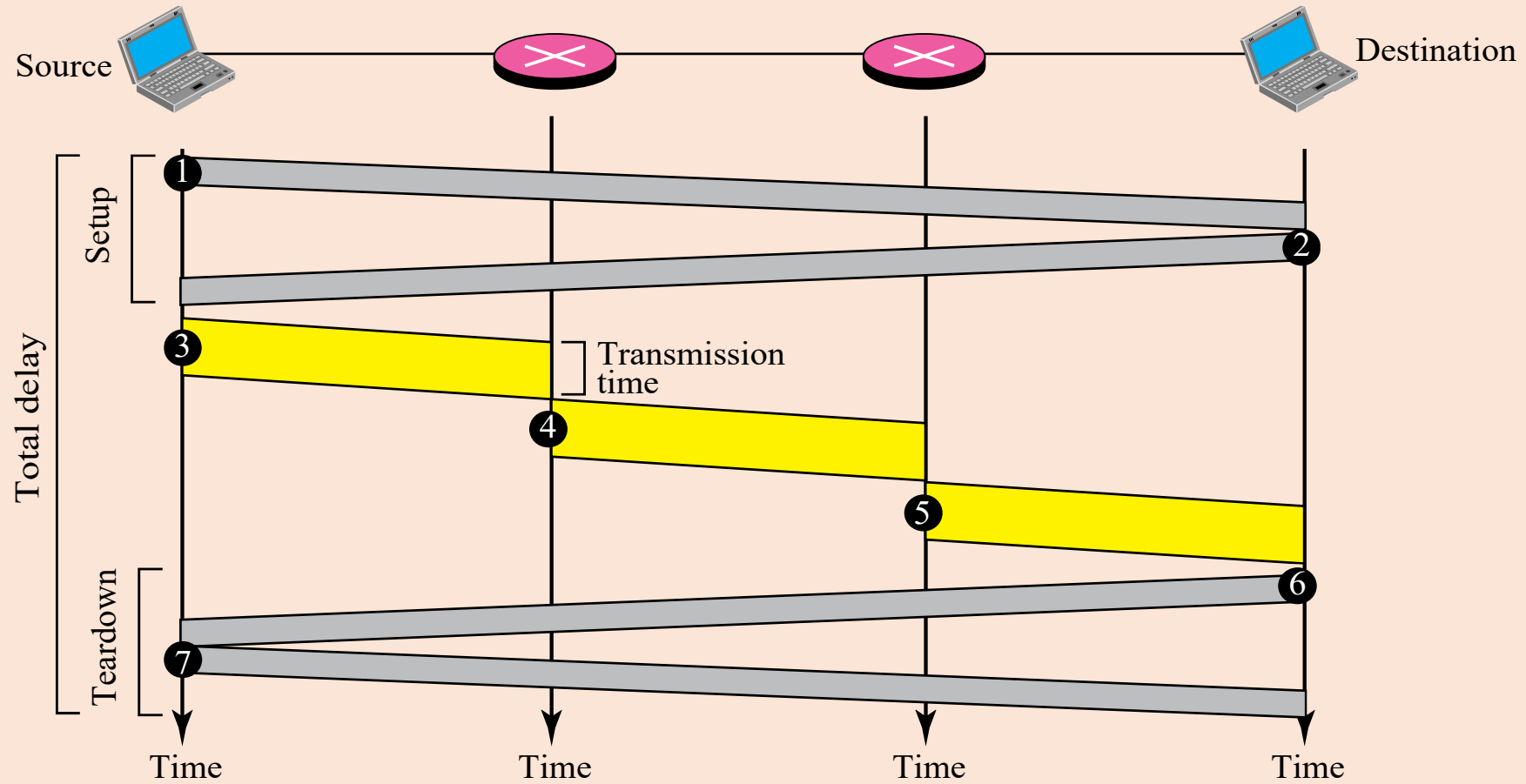
After sending all packets from A to B

A sends a special packet called a **teardown packet**.

Destination B responds with a **confirmation packet**.

All routers **delete the corresponding entry** from their tables.

Delay in Connection Oriented Network



NETWORK LAYER SERVICES

What are **the services** provided by the network layer ?

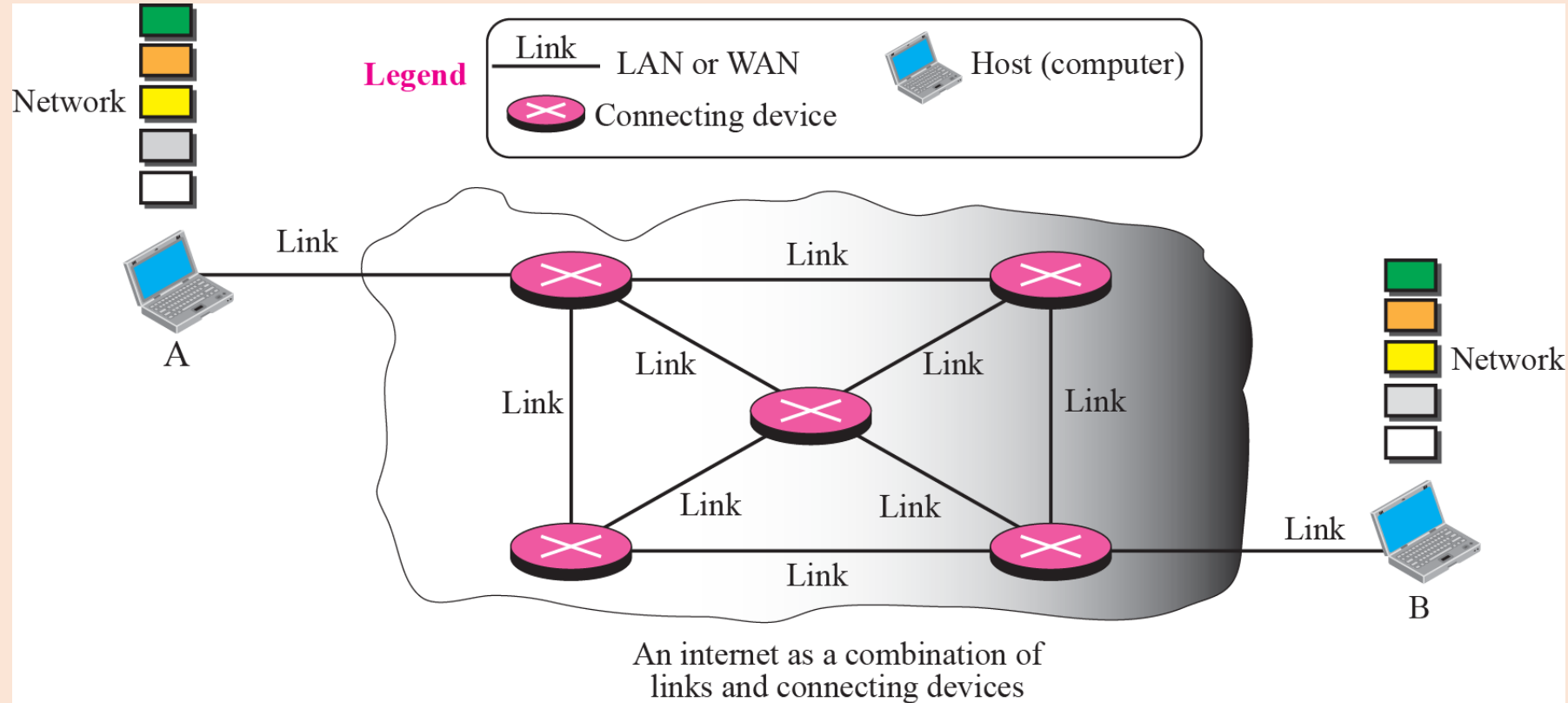
Our discussion will be based on the **connectionless service**, the **dominant service** in today's Internet.

Multiprotocol Label Switching (MPLS) is a routing technique in telecommunications networks that directs data from one node to the next based on short path labels rather than long network addresses(IP).

IPv6 is being introduced to overcome the address shortage of the current IPv4 protocol, but it also offers a new feature, i.e., the **Flow Label** field in the IPv6 packet header. However, it is used very little in practice

NETWORK LAYER SERVICES

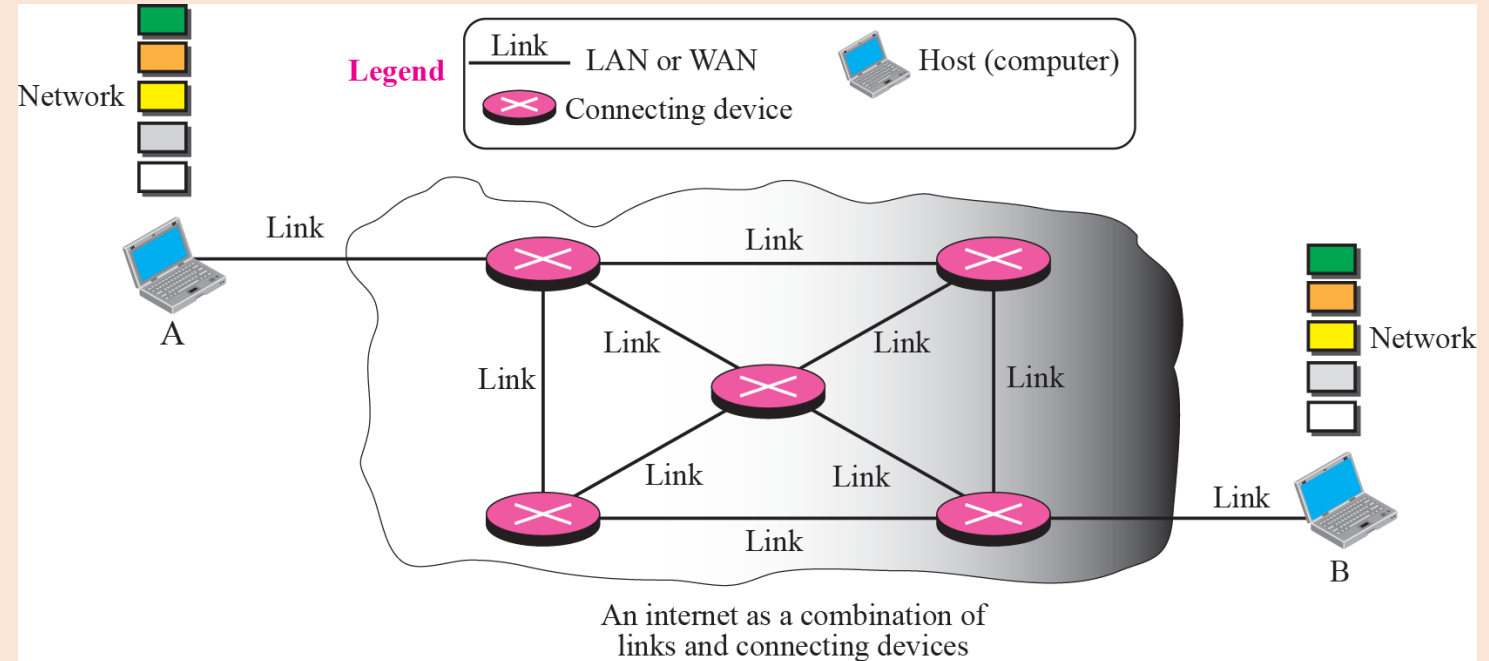
Why Network Layer ?



NETWORK LAYER SERVICES

Services provided.

- a. Logical Addressing
- b. Services Provided at the Source Computer
- c. Services Provides at the Each Router
- d. Services Provided at the Destination Computer

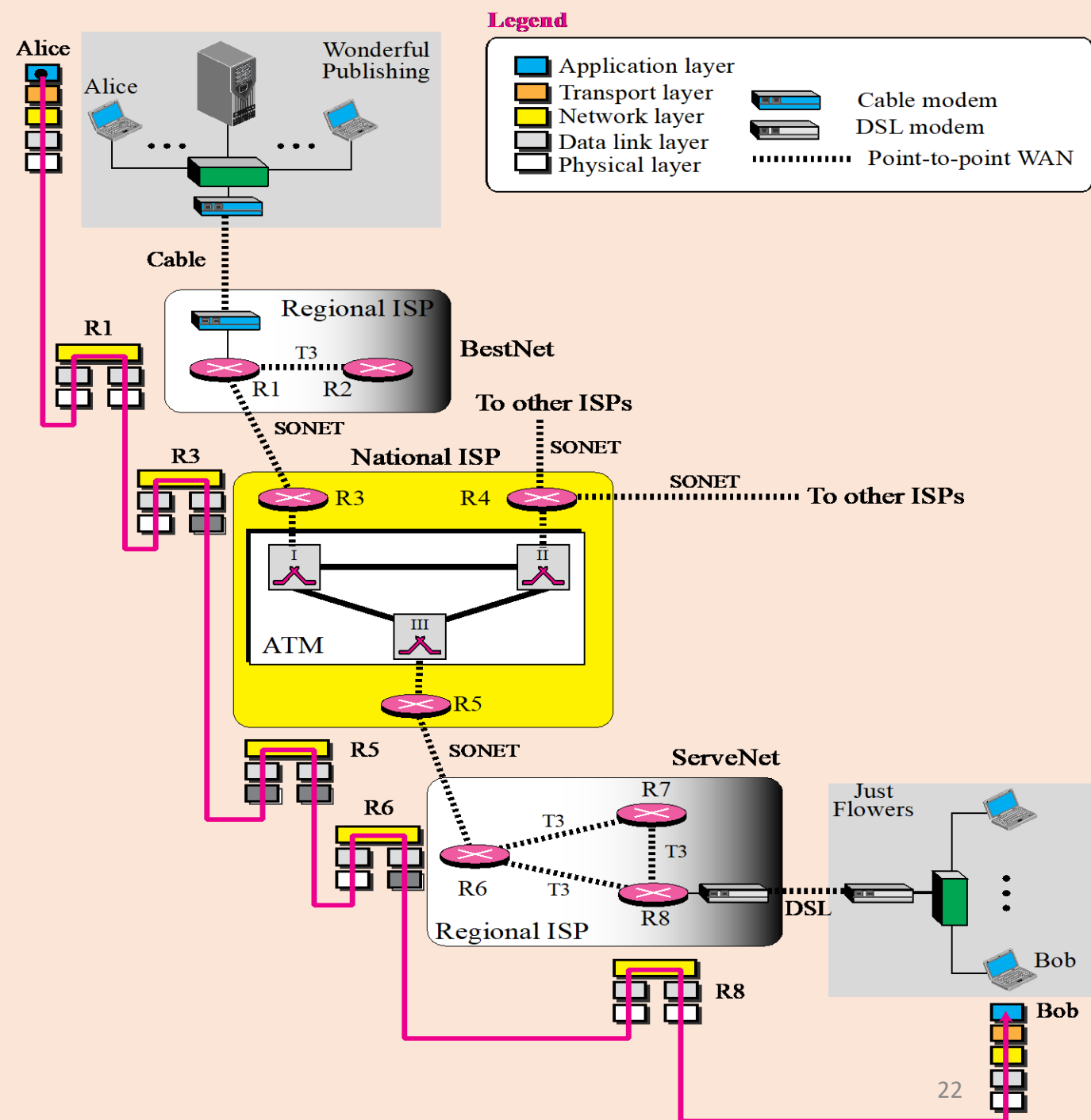


NETWORK LAYER SERVICES

Imaginary part of the Internet & TCP/IP Layers

The **two computers** are involved in **five layers**;

The **routers** are involved in **three layers** of the TCP/IP protocol suite.

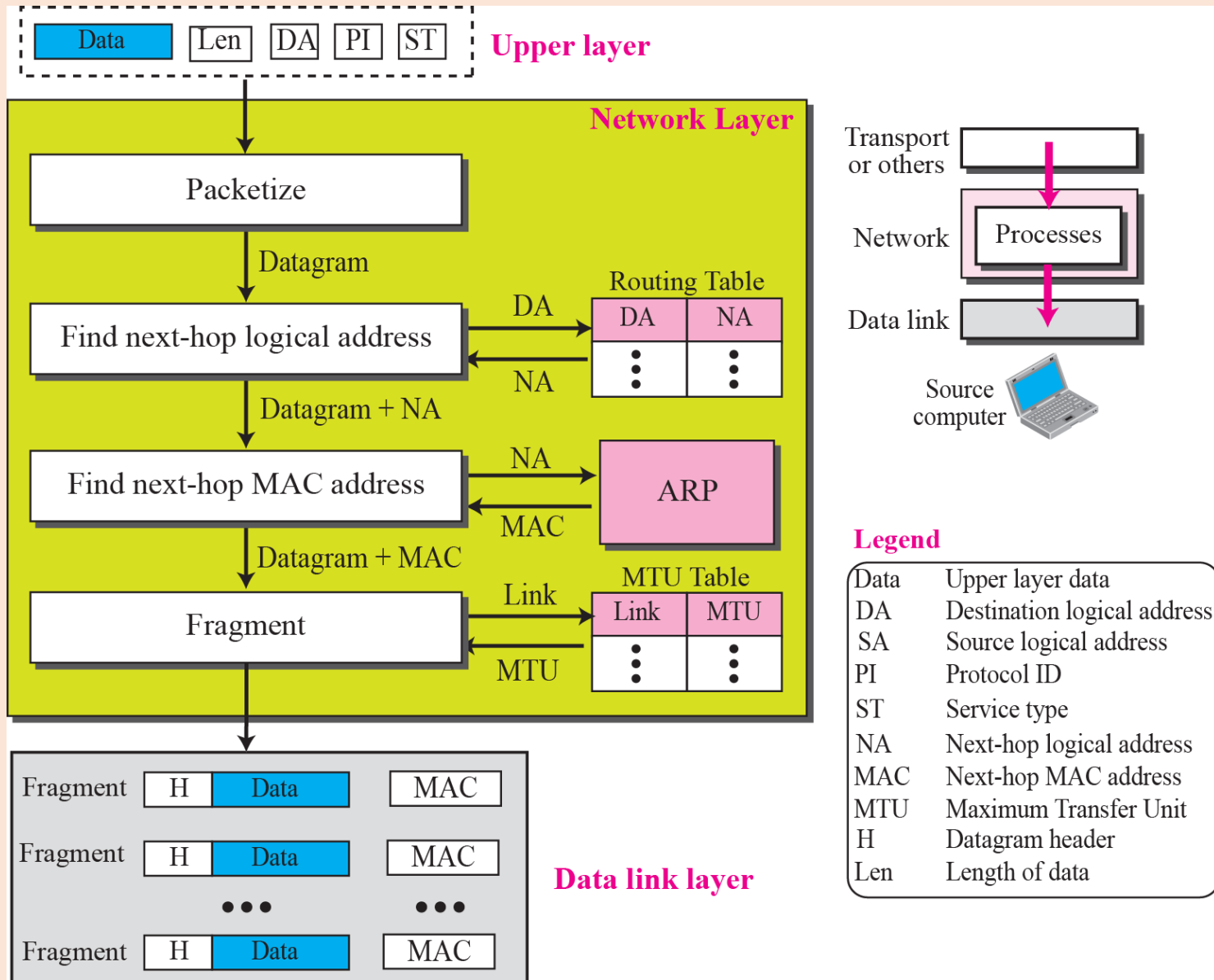


a. Logical Addressing

- The network layer provides end-to-end communication.
- Each system in the network needs universal identification system(Logical Address/ Network Layer Address).
- IPv4 is the addressing scheme used(32 bit address)
- New one is IPv6 addressing scheme

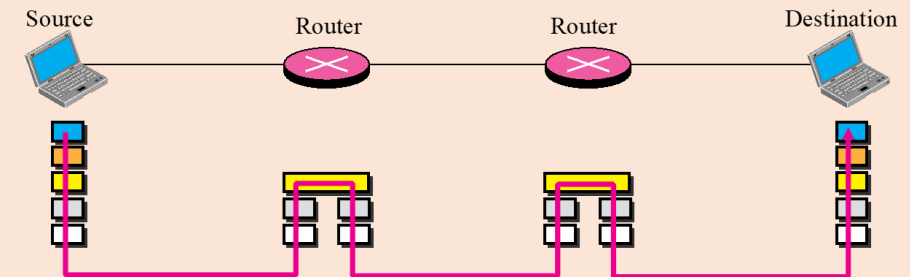
NETWORK LAYER SERVICES

b. Services Provided at the Source Computer



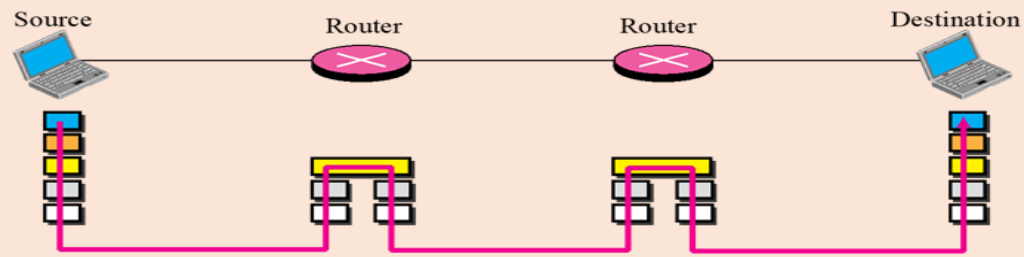
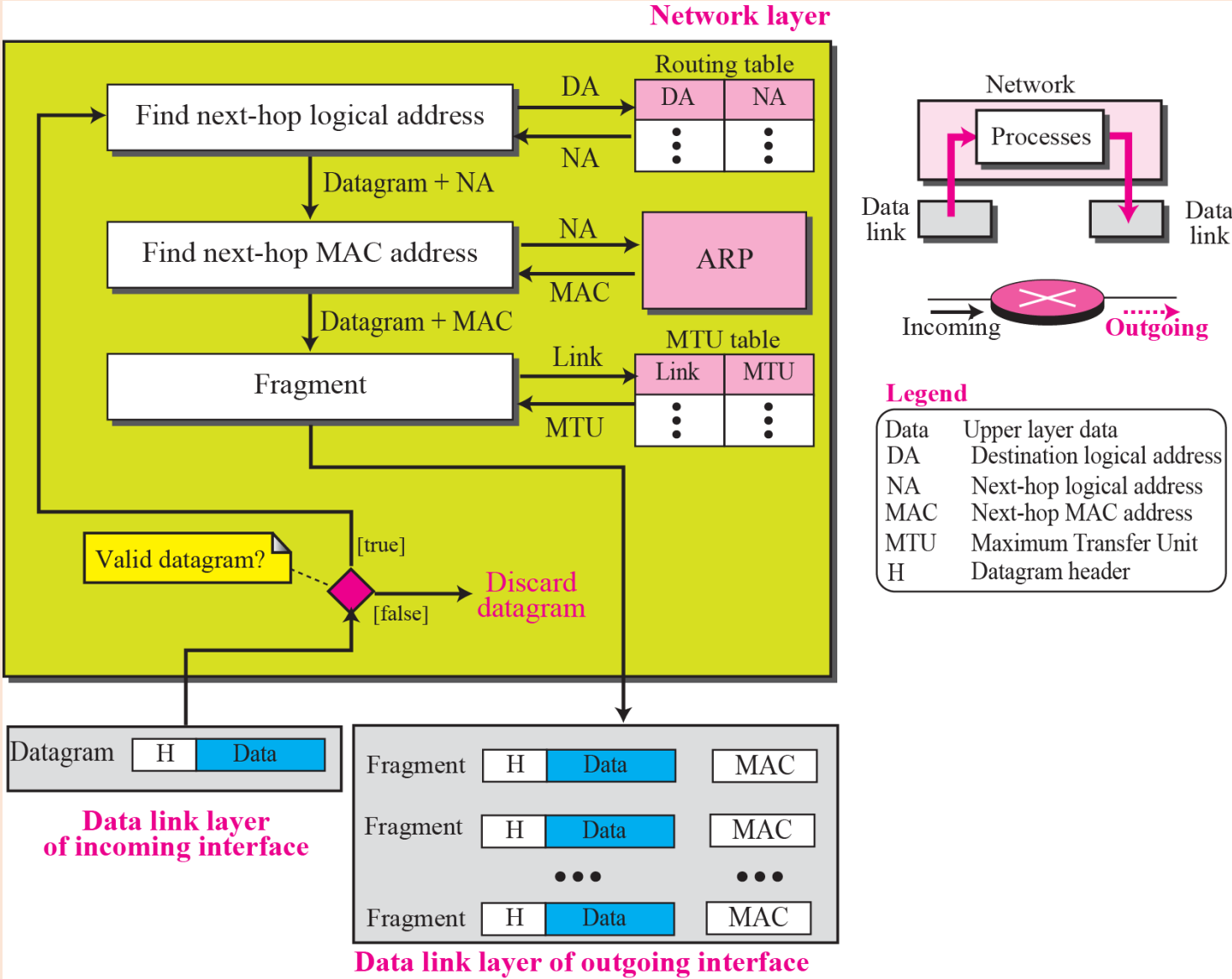
The network layer at the source computer provides **4 services**:

- ✓ **Packetizing**
- ✓ Finding the **logical address** of the next hop
- ✓ Finding the **physical (MAC) address** of the next hop
- ✓ **Fragmenting**, the datagram if necessary.



NETWORK LAYER SERVICES

C. Services Provides at the Each Router



Validates incoming Datagram.

The network layer at the Router provides **3 services**:

- ✓ Finding the **logical address** of the next hop
- ✓ Finding the **physical (MAC) address** of the next hop
- ✓ **Fragmenting**, the datagram if necessary.

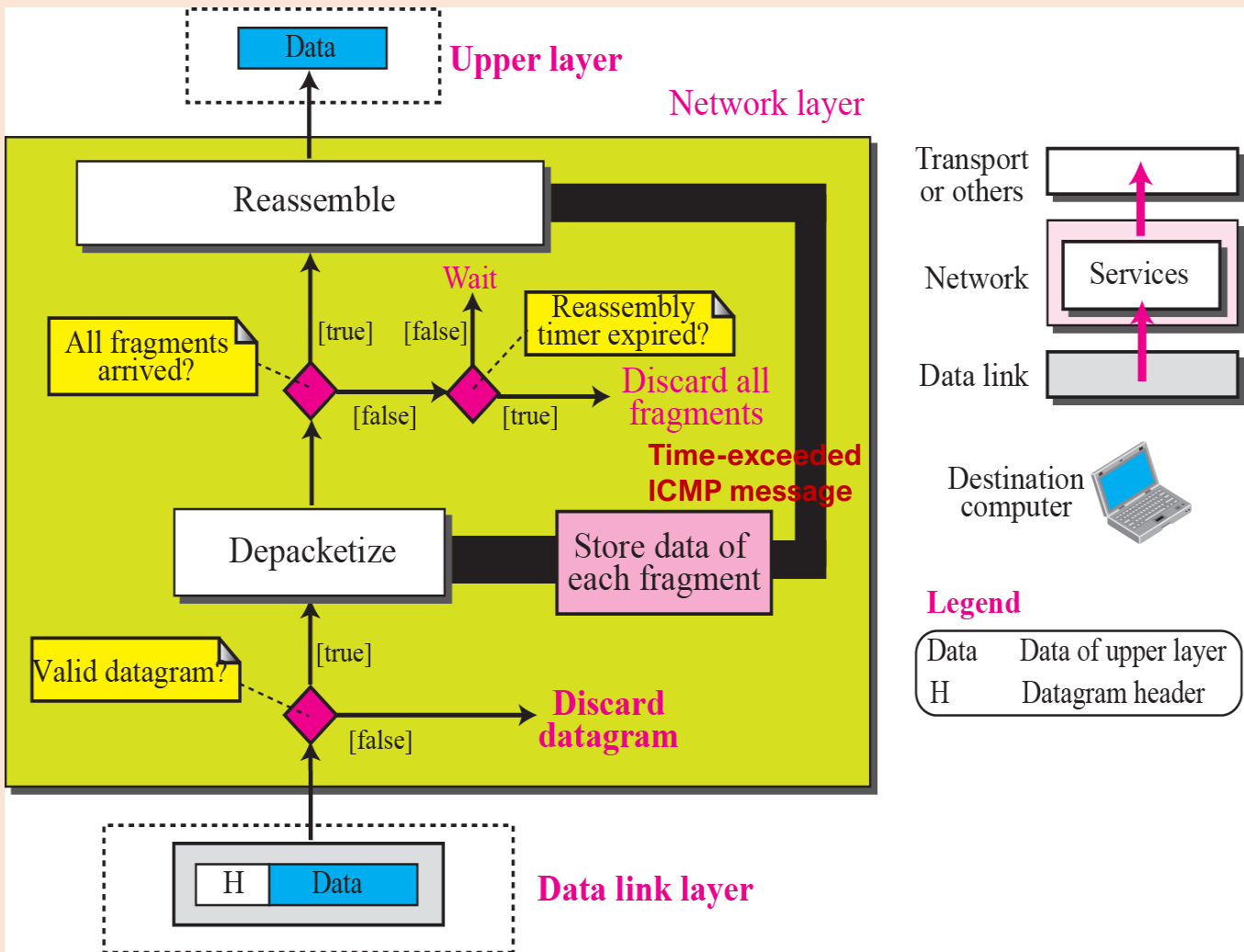
Recalculates Network Packet header checksum & adds to the header

Router, therefore, needs **to interact** with **two data link layers(DLL)**

DLL of **incoming** interface
DLL of **outgoing** interface

NETWORK LAYER SERVICES

d. Services Provided at the Destination Computer



Validates incoming Datagram.

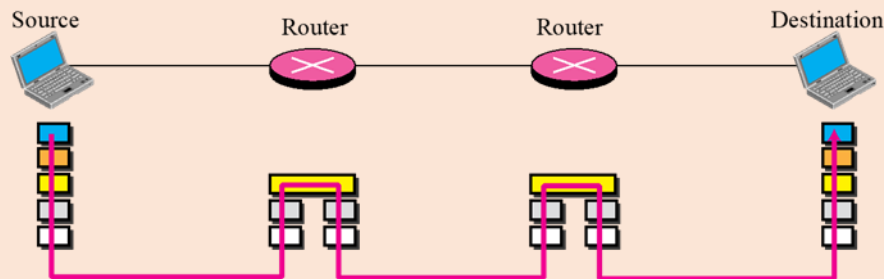
Depacketize.

Store received **fragment** (**reassembly Timer** started)

If All Fragments not arrive before timer expiry then
Destroy received fragment.

Else

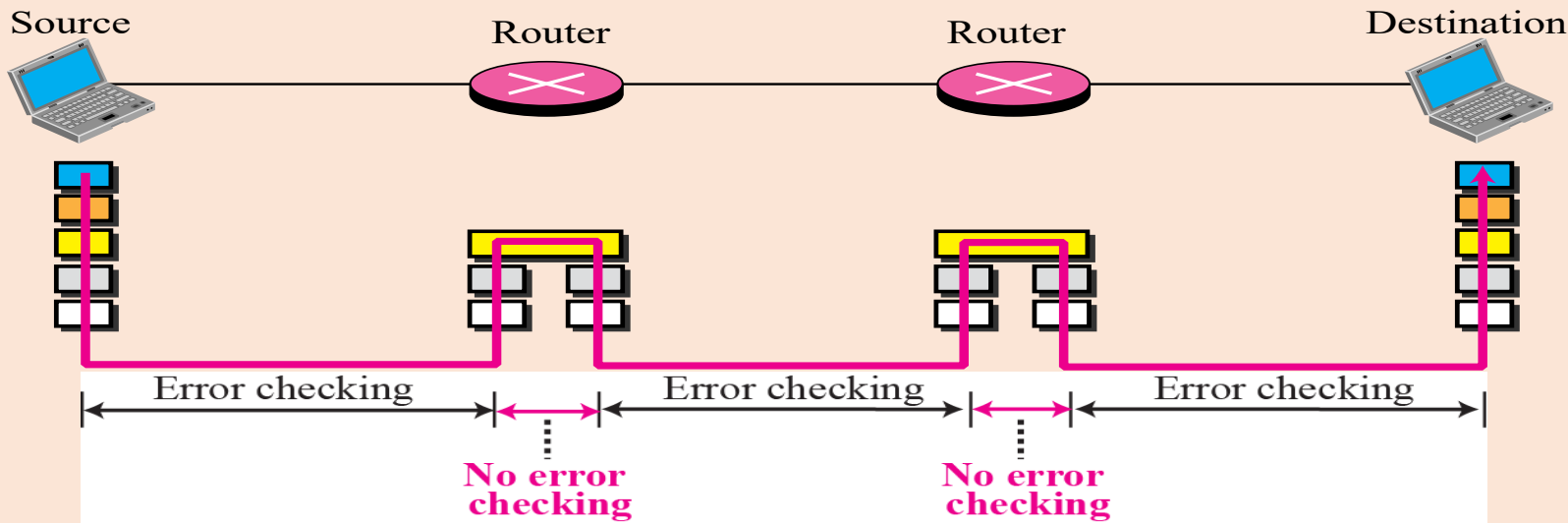
Reassemble fragments in Order



OTHER SERVICES

–**Error control** : A mechanism for detecting corrupted, lost, or duplicate Datagrams.

Since a **hop-to-hop error control (CRC)** is already implemented at the **data link layer**, why do we need error control at the network layer?



The data link layer can miss any error that occurs when the datagram is being processed by the router. To keep Network layer processing fast, Network Layer do not implement rigorous error checking. However, N/W layer **header checksum** is used – to prevent packet being delivered to wrong destination. The network layer at the Internet does not directly provide error control, the Internet uses another protocol- **ICMP** (Internet Message Control Protocol) to report error situation to the Sender.

Flow control

Regulates the amount of data a source can send without overwhelming the receiver.

- The network layer in the Internet, however, does not directly provide any flow control.

The datagrams are sent by the sender when they are ready without any attention to the readiness of the receiver.

- Flow control responsibility is given to the upper layer protocols (Transport layer) which use N/W layer protocol.

Congestion control

Congestion in the network layer is a situation in which too many datagrams are present in an area of the Internet.

– Congestion in a connectionless network can also be implemented using a **choke Packet**.

The network layer uses an auxiliary protocol, **ICMP** to inform Sender to slow down .

– Another way is to rank packets based on importance.

- In a connection-oriented network

- One method simply creates an extra virtual circuit.
- A better solution is advanced negotiation during the setup phase.

Other services are – Routing , QoS (upper layer) and Security

END